CSA 1305 192371005

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TASK-1 Regular Expression Design.

i) Binary strings ending with 01

$(0+1)^* 01$

ii) Binary strings containing substring 11 or 00

$(0+1)^* (00+11)(0+1)^*$

iii) Strings over containing at least 1 occurence of ab

$(a+b)^* ab(a+b)^*$

iv) Strings over 1 having an even no. of zero

$1^* (01^* 01^*)^*$

v) Strings over starting with a and ending with b

$a(a+b)^* b$

TASK-2

Gn. regular expression:

$R = (0+1)^* 101 (0+1)^*$

The regular expression represents all binary strings containing the substring 101 at least once. Any combination of 0s and 1s can appear before or after the substring 101.

Alphabet: $\Sigma = \{0, 1\}$

Substring: 101

Five accepted strings

101, 0101, 1010, 11010, 101101

Three rejected strings

000, 111, 1001

Role of Kleene Star and Union.

Kleene Star (*): Represents zero or more repetitions of 0s and 1s. It allows any binary sequence before and after 101.

Union (+): Represents OR. Here, (0+1) means either 0 or 1 can occur.

TASK-3

i) $L_1 = \{a^n b^n \mid n \geq 0\}$

$S \rightarrow aSb \mid \varepsilon$

Start symbol: S

Non Terminals: S

Terminals: a, b

Ex: ε, ab, aabb

ii) $L_2 = \{ a^n b^m \mid n, m \geq 0 \}$

$S \rightarrow AB$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid \varepsilon$

Start symbol : S

Non-terminals : A, B

Terminals : a, b

Example : ε, a, b, ab, aab, abb

iii) Arithmetic expressions involving identifiers and addition (+)

$E \rightarrow E + id \mid id$

iv) Balanced parentheses

$S \rightarrow (S)S \mid \varepsilon$

v) Binary strings containing equal nos. of zeros & ones

$S \rightarrow 0S1S \mid 1S0S \mid \varepsilon$

Ex: ε, 01, 10, 0011, 0101

TASK-4

Gn. Content-Free grammar: $S \rightarrow aSb \mid \varepsilon$

Start symbol : S

Non terminals : S

Terminals : a, b

ii) Generate strings for $n = 0, 1, 2, 3$

n	String
0	ε
1	ab
2	aabb
3	aaabbb

iii) Leftmost derivation of aaabbb

$S \rightarrow aSb$

$\Downarrow$

$aaSbb$

$\Downarrow$

$aaaSbbb$

$\Downarrow$

ε

$\Rightarrow$ aaabbb

iv) Rightmost derivation of aaabbb

$S \rightarrow aSb \mid \varepsilon$

$S \rightarrow aSb$

$\Downarrow$

$aaSbb$

$\Downarrow$

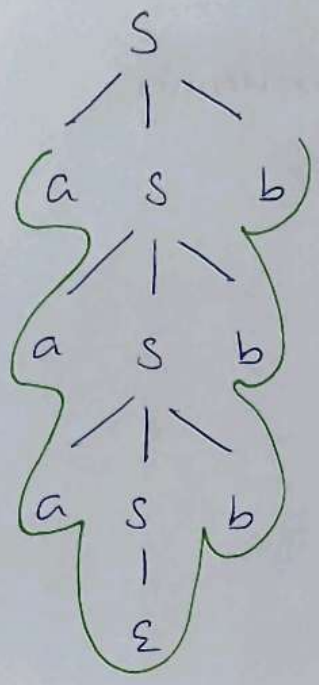
$aaaSbbb$

$\Downarrow$

ε

$\Rightarrow$ aaabbb

x) Parse tree for aaabbb



String : aaa bbb

TASK-5

i) Production Rules

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid id$

ii) Grammar components

Start symbol : E

Non terminals: { E, T, F }

Terminals: { id, +, *, (,) }

iii) Operator Precedence

Multiplication (*) has higher precedence than addition (+)

Parantheses are used to group expressions

iv) Leftmost derivation of a+b*c

$E \rightarrow \underline{E} + T$

$\rightarrow \underline{T} + T$

$\rightarrow \underline{F} + T$

$\rightarrow id + \underline{T}$

$\rightarrow id + \underline{T} * F$

$\rightarrow id + \underline{F} * F$

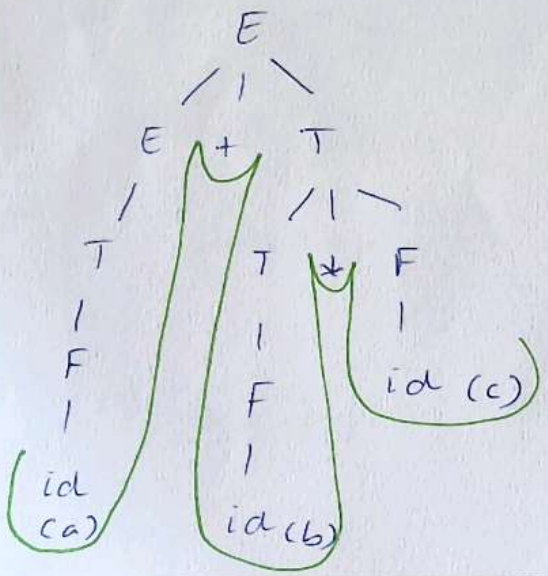
$\rightarrow id + id * \underline{F}$

$\rightarrow$ **id + id * id**

Replacing with a, b, c

a+b*c

v) Parse tree for a+b*c



a+b*c